

Melody Yin (No Sponsorship Required)

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Education

Carnegie Mellon University

MS in Electrical and Computer Engineering

Expected May 2027

University of California, San Diego

BS in Math-Computer Science; BS in Cognitive Science, Specialization in Machine Learning

Sept 2021 – Jun 2025

GPA: 3.8/4.0 | Provost Honors (6x), PACE Fellowships '23, '24

Work Experience

Software Engineering Intern – HU MEI USA INC – Rancho Cucamonga, CA

Jun 2025 – Sept 2025

- Developed an AI-powered travel assistant that supports natural language interaction in both text and voice, using **LangChain**, **FastAPI**, and **WebSockets**. Integrated AI-based text-to-speech (**TTS**) to enable real-time conversational guidance for clients throughout their travel planning process.
- Implemented long-term memory with **Redis** + **LangChain** to persist conversation state, improving multi-turn coherence and contextual consistency by **45%** for flight, hotel, and itinerary recommendations.
- Built a **RAG pipeline** using **LangChain embeddings** + **Qdrant** to retrieve context-relevant data, reducing hallucination rates by **35%** and improving accuracy in complex travel queries (visa policies, multi-leg itineraries, booking comparisons).
- Integrated **SerpApi** for web search aggregation with internal itinerary data, applying ranking and filtering to deliver the most relevant travel insights and consulting recommendations.

Software Engineering Intern – ETL Performance INC – Remote

Jun 2024 – Sept 2024

- Developed scalable **microservices** using **Golang (GIN)** with **event-driven** architecture for ETL's testing platforms, supporting real-time sensor data ingestion, test logging, configuration management, and anomaly keyword extraction in cooling system validation.
- Implemented **gRPC** with connection pooling and keepalive strategies to enable low-latency, high-throughput data exchange across R&D and dual-environment testing workflows.
- Designed and optimized **MySQL** schemas to persist test records and performance logs, applying indexing and query caching to accelerate analytics queries for engineering teams.
- Integrated **Redis** caching with an LRU policy to speed up access to frequently used test configurations, sensor streams, and recent test histories, resulting in a **23%** reduction in query latency during engineering simulations.
- Configured **Nginx** for reverse proxy and load balancing, and containerized services with **Docker** deployed via **Docker Swarm**, enhancing ETL's distributed validation pipeline scalability and fault tolerance.

Data Analyst Intern – UC San Diego Career Center – San Diego, CA

Jan 2024 – Jun 2025

- Analyzed UCSD Career Center's advising data using **Python (Pandas, NumPy)** for feature engineering, cleaning and transformation; automated **ETL** pipelines reduced manual effort and enabled repeatable reporting workflows.
- Visualized trends with **Matplotlib/Seaborn**, applied statistical analysis, and provided staffing optimization insights that improved peak-hour coverage by **20%**.

Projects

Distributed RPC Framework

- Developed MyRPC to mimic the feature in **gRPC**, using **protobuf** as the interface description language, supporting asynchronous calls, dynamic service registration, discovery, and heartbeat detection with **Spring**, **Netty** and **Zookeeper**.
- Implemented long-lived TCP connections, heartbeat detection, and synchronous I/O operations using **Netty**.
- Enabled service discovery, service registration, **load balancing** with weighted round-robin, and health monitoring with **Zookeeper**.
- Integrated JSON serialization and de-serialization capabilities for efficient and fast data transmission over network using **FastJSON**; dynamically generated proxy classes for remote service interfaces using **CGLIB**.

Distributed Key-Value Storage System

- Built a distributed key-value store in **Java** with **Mmulti-Raft** consensus, achieving **99.9%** availability, 50K QPS, and <10ms P99 latency for linearizable reads/writes.
- Implemented **consistent hashing** with virtual nodes for **95%** balanced data distribution and dynamic shard rebalancing with <1% performance overhead.
- Integrated **RocksDB (LSM-tree)** + in-memory B+ Tree hybrid engine with **MVCC** snapshot isolation; optimized reads via **ReadIndex**, **FollowerRead**, and **async Apply** batching, reducing latency by **35%** and doubling throughput.

Skills

Programming Languages: Java, Python, C, Golang, C++, JavaScript, TypeScript

Frameworks & Tools: Spring Boot, Spring Cloud, Spring Security, Node.js, Express.js, React.js, Maven, JUnit, Mockito, AWS (EC2, S3, Lambda), Docker, Kubernetes, Jenkins, CI/CD pipelines, MySQL, MongoDB, Redis, PostgreSQL, Elasticsearch, RocketMQ, Apache Kafka, RabbitMQ, Git, Linux, Shell scripting, Swagger, Postman